

Does Syntactic Structure Make Language Models More Robust to Noise?

LSTM vs. RNN Subject-Verb Agreement Under Training-Data Corruption

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Outline

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Motivation: Can grammar survive noisy input?

- Children acquire robust grammatical systems despite hearing noisy and imperfect language
- This observation motivates the classic **Poverty of the Stimulus** debate:
Is grammar learned from data, or supported by innate structure?
- Human experiments cannot systematically manipulate children's linguistic input
- Neural language models provide a controllable testbed:
 - architecture = innate structure
 - training corpus = linguistic environment

RQ1: Can grammar be learned from noisy input?

- How much grammatical corruption can a model tolerate?
- Is there a critical noise threshold beyond which acquisition breaks down?

RQ2: How does noise change generalization?

- Does the model maintain a hierarchical syntactic rule?
- Or does it fall back on a simpler linear nearest-noun heuristic?

RQ3: Does structure buy robustness?

- Compare LSTM and RNNG under identical noise conditions.
- Does explicit syntactic bias help preserve grammatical learning?

Subject-verb agreement: hierarchical vs. linear

- **Hierarchical rule:** agree with the true subject.
- **Linear rule:** agree with the noun closest to the verb.

Example

The keys to the cabinet _____ on the table.

- True subject = *keys*
- Nearest noun = *cabinet*
- Hierarchical rule $\Rightarrow$ **are**
- Linear heuristic $\Rightarrow$ **is**

Key idea: SVA creates cases where the two rules make different predictions.

Attractors as the diagnostic case

- Both rules make different predictions.
- Success requires ignoring misleading linear cues.
- Attractor accuracy reveals the model's generalization strategy.

Two competing learners

LSTM

- Sequential recurrent model
- Processes sentences word by word
- No explicit syntactic structure
- Learns grammar from statistical regularities

Weak structural prior

RNNG

- Generates sentences and parse trees jointly
- Explicit hierarchical representations
- Encodes phrase structure directly
- Strong syntactic inductive bias

Strong structural prior

Key Question

Does explicit syntactic structure help preserve hierarchical grammatical rules under noisy input?

SVA as a syntax benchmark

- Linzen (2016): LSTMs learn agreement but struggle with attractors.
- Gulordava (2018): evidence for abstract syntactic generalization.

Structural inductive bias

- Dyer (2016): Recurrent Neural Network Grammar (RNNG).
- Kuncoro (2018): explicit syntax improves long-distance dependencies.
- Prior work studies noisy inputs and grammatical corruption.
- Focus is typically on downstream performance.

Research gap

- Little work asks whether grammatical rules themselves can still be acquired under noisy supervision.

- No systematic study of how SVA corruption affects:
 - rule acquisition,
 - hierarchical vs. linear generalization,
 - LSTM vs. RNNG robustness.

Noise injection: five corrupted corpora

- Base: Linzen et al. (2016) SVA corpus — $\sim 1.58\text{M}$ Wikipedia sentences, vocab capped at the 10K most frequent types.
- Each sentence annotated with subject, verb, verb position, and # intervening attractor nouns.
- Build five training corpora, identical except agreement noise: flip the verb's number with probability p .

$$p \in \{0, 0.004, 0.02, 0.10, 0.50\} \quad (\text{baseline} \rightarrow \text{high})$$

- Flip via deterministic inflection (lemminflect + irregular copulas/auxiliaries); all other lexical/syntactic content preserved.
- Training-set size held constant $\Rightarrow$ isolates noise.

Leak-free evaluation protocol

- The five corpora are row-aligned with the source corpus.
- Reserve one fixed held-out set: 20k sentence indices (seed 1234).
- Those rows excluded from **every** training corpus $\Rightarrow$ no test leakage under any noise condition.
- Test items built from the **uncorrupted** source $\Rightarrow$ all test sentences are grammatical.
- Same test set for all conditions — only training noise varies.

- For each held-out sentence: a prefix, the grammatical verb v^+ , and its number-flipped counterpart v^- .
- Model is **correct** iff it prefers the grammatical form:

$$P(v^+ \mid \text{prefix}) > P(v^- \mid \text{prefix})$$

- Each pair tagged `has_attractor`: an intervening noun mismatching the subject in number.

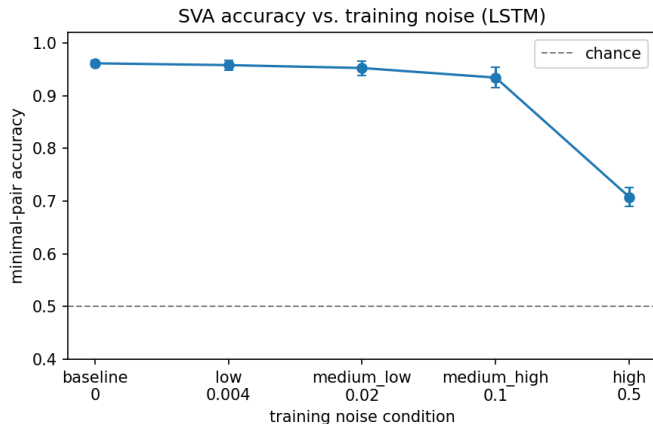
- **Sequential baseline** — word-level LSTM LM (Linzen 2016 / Gulordava 2018 style): 2 layers, tied embeddings, dropout; trained from scratch per condition.
- **Structurally-biased** — RNNG (Dyer et al. 2016): explicit hierarchical structure.
- Core question: does a structural bias make hierarchical agreement *more robust* to noisy supervision?
- Fallback: ON-LSTM (Shen et al. 2019) — softer structural bias, no parse supervision needed.

- Primary: minimal-pair **accuracy** (grammatical $>$ ungrammatical).
- Secondary (graded): **surprisal** of the correct verb, $s = -\log_2 P(v^+ \mid \text{prefix})$.
- Split accuracy by attractor presence; reliance on the linear cue = **attractor gap**

$$\Delta = \text{acc}_{\text{no-attr}} - \text{acc}_{\text{attr}}$$

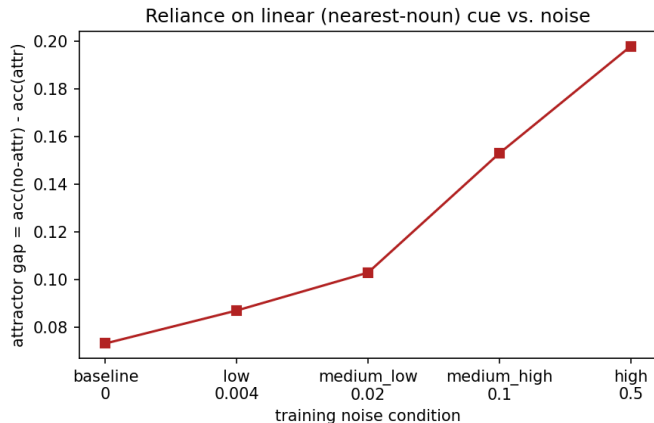
- Small $\Delta \rightarrow$ hierarchical rule; large $\Delta \rightarrow$ nearest-noun (linear) rule.
- Averaged over seeds $\{1, 2, 3\}$; central analysis: Δ vs. noise.

Accuracy degrades with noise — with a threshold



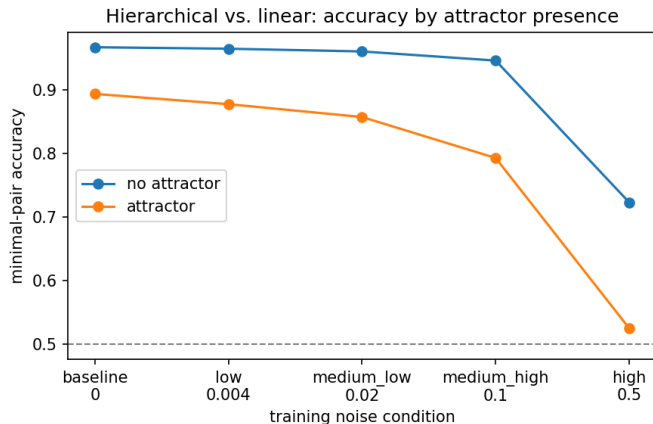
- Flat up to 10% noise: 0.961 $\rightarrow$ 0.934
 - Collapses at 50%: down to 0.708
 - Tolerates ~ 1 bad verb in 10, then breaks
- $\Rightarrow$ critical-threshold effect, not gradual decay

The attractor gap widens with noise



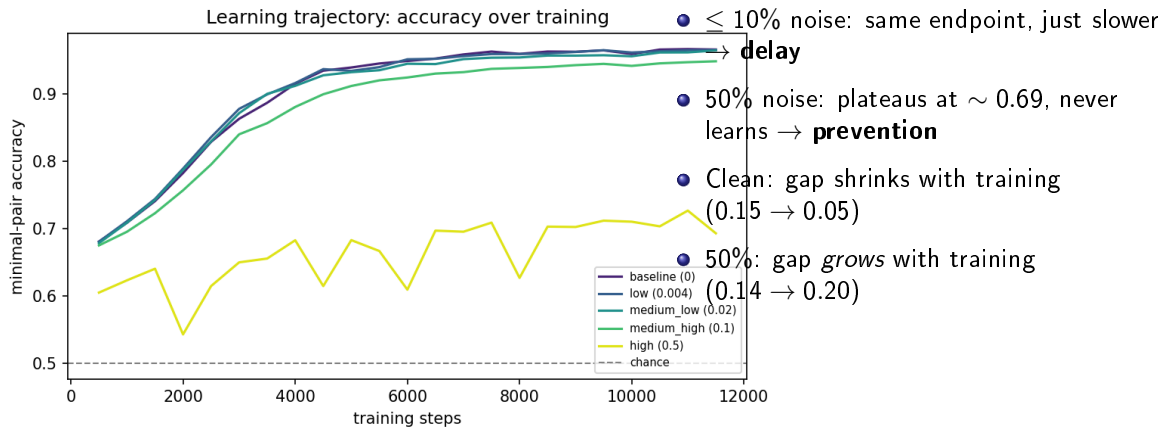
- Gap = $\text{acc}(\text{no-attractor}) - \text{acc}(\text{attractor})$
 - Grows monotonically: $0.073 \rightarrow 0.198$
 - Index of reliance on the *linear* nearest-noun rule
- ⇒ noise pushes the model toward the shortcut

Accuracy by attractor presence



- No-attractor items stay > 0.94 until 50%
 - Attractor items fall to near chance (0.525)
 - The two rules disagree only on attractor items
- ⇒ that is exactly where noise bites first

Learning trajectories: delay vs. prevention



When does the rule break?

- **Threshold, not gradual:** absorbs noise up to 10%, snaps at 50% — like a child's grammar surviving occasional speech errors.
- **Attractor gap is the early warning:** it climbs ($0.07 \rightarrow 0.20$) while overall accuracy still looks healthy.
- **Delay vs. prevention:** low noise only *delays* acquisition (curves converge); 50% noise *prevents* it and entrenches the linear rule.

Does structure buy robustness?

- The headline question — does a syntactic prior (RNNG) resist noise better than a sequential LSTM? — needs the RNNG runs (in progress).
- **LSTM baseline:** no explicit hierarchy bias $\Rightarrow$ learns the rule cleanly when input is clean, but drifts to the linear shortcut as noise grows.
- If RNNG holds the hierarchical rule where the LSTM has caved $\Rightarrow$ structure buys robustness.
- If both fail together $\Rightarrow$ the bottleneck is input quality, not architecture — speaking to poverty-of-the-stimulus.

Takeaways

- LSTMs tolerate moderate SVA noise but have a sharp breaking point.
- Noise trades the hierarchical rule for the linear nearest-noun heuristic.
- Heavy noise prevents acquisition outright — more training makes it worse.

Limitations & next steps

- Synthetic, uniform noise; templatic corpus; single-seed trajectories.
- Complete the RNNG arm; try ON-LSTM as a no-gold-tree syntactic baseline.
- Move toward naturalistic, frequency-dependent noise and larger scale.

Thank you!

Questions?